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FAIRNESS IN HEALTHCARE E-ROSTERING: AN EMPIRICAL ANALYSIS OF ANTECEDENTS AND EFFECTS

Completed Research Paper

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Abstract

Fairness in healthcare shift scheduling is crucial for well-being of workers but remains challenging to achieve due to numerous requirements. While electronic rostering (e-rostering) helps manage the complexity of shift scheduling, its impact on fairness remains unclear. First, we develop a stimulus-organism-response model guided by organizational justice theory to explain how contextual factors in healthcare shift scheduling (schedule predictability, scheduling control, and relationship with supervisors) shape workers' scheduling fairness perceptions and, in turn, influence job satisfaction. Second, we examine how e-rostering, compared to human scheduling, alters the relationships in this model. Using data from 185 healthcare workers, our results show that contextual factors significantly impact fairness perceptions, and salience shifts from procedural to distributive fairness in determining job satisfaction under e-rostering. We contribute to future research by highlighting how technologies can reshape the relationship between fairness and work attitudes, and to practice by offering design guidance for fair e-rostering.

Keywords: organizational justice theory, electronic rostering, healthcare workers, healthcare shift scheduling, stimulus-organism-response

1 Introduction

Personnel shortages and high turnover rates are critical and persistent challenges in the healthcare sector. When the demand for skilled healthcare workers is unmet, patient care quality declines, and the work pressure on the remaining staff intensifies (De Bienassis & Lafortune, 2024; Kumar et al., 2025; OECD, 2025). A key driver of turnover in the healthcare sector is low job satisfaction (Brewer et al., 2015). Therefore, improving job satisfaction is a central pathway to retaining healthcare workers and attracting new junior staff as well as career changers from other sectors (OECD, 2025). Shift scheduling, also known as rostering, is an essential determinant of job satisfaction because it determines working hours. Unlike a regular 9-to-5 workday, healthcare facilities often provide services around the clock, which require healthcare workers to be available beyond weekdays, including nights, weekends, and holidays. Thus, shift schedules have a direct impact on private lives and the well-being of healthcare workers, making fairness an essential factor in scheduling (Uhde et al., 2020). For example, fairness in shift scheduling entails allocating shifts within a ward while accounting for essential needs of individual workers, such as leisure activities, care of dependents, and health conditions (Epstein et al., 2023).

Still, creating fair shift schedules requires integrating organizational, technical, and social aspects. From an organizational perspective, scheduling is a demanding task, owing to numerous strict operational and legal requirements to guarantee the quality of care. Technical solutions, referred to as electronic rostering (e-rostering), have been increasingly used to more efficiently account for requirements as they offer higher processing capacity than human scheduling (O'Connell et al., 2024; Schlicker et al., 2021). However, current e-rostering primarily focuses on efficient schedule generation but seldom considers workers' perspectives on fairness or how it is shaped by contextual factors. This is partly because scheduling policies of healthcare facilities, which underpin e-rostering, lack a clear, actionable definition of fairness (Fuchs et al., 2025; Petrovic, 2019). Moreover, even when e-rostering features are intended to ensure fairness, they primarily reflect developers' notions of fairness, which often diverge from recent research on worker-centered shift scheduling in healthcare (Bieri & Matt, 2025; Erhard et al., 2018). Although this recent research has identified relevant aspects of fairness and its antecedents, such as contextual factors, it does not yet integrate these aspects conceptually or explain how they inform e-rostering design (e.g. Barker et al., 2025). This gap is critical because if workers' perspectives on how fair schedules should be generated are not incorporated, e-rostering may yield allegedly "fair" shift schedules that workers ultimately reject, which jeopardizes the acceptance of e-rostering (Dolata et al., 2022; Erhard et al., 2018). Accordingly, a conceptual and empirical understanding is required to examine how e-rostering affects the perceptual basis of fairness and contextual factors among healthcare workers, thereby supporting the integration of these aspects into e-rostering (Fuchs et al., 2025).

To address this gap, we developed a conceptual model that integrates healthcare workers' perceptions of scheduling fairness, contextual factors in shift scheduling, and job satisfaction as key work attitude to explain turnover in the healthcare sector, to examine how e-rostering affects the perceptual basis of workers. Building on Ochmann et al. (2024), who integrated organizational justice theory (OJT) into the stimulus-organism-response (S-O-R) framework, we conceptualize a model in which three central contextual factors of shift scheduling (schedule predictability, scheduling control, and relationships of workers with supervisors) shape the "organism," operationalized by perceived fairness regarding final schedules (i.e., distributive fairness) and scheduling procedures (i.e., procedural fairness). We link scheduling fairness to behavioral responses in the form of job satisfaction. We also theorize how these relations differ when e-rostering is employed as a stimulus rather than human scheduling. Our analysis of data from 185 healthcare workers shows that under e-rostering, scheduling control exerts a stronger impact on scheduling fairness, whereas the exchange with supervisors remains equally relevant. Further, we observed a shift in salience from procedural to distributive fairness when shaping job satisfaction. For future research, we suggest that emerging technologies, such as e-rostering, can reshape the relationship between fairness perceptions and work attitudes (Colquitt, 2012). Additionally, incorporating contextual factors into the integration of OJT and the S-O-R framework helps explain perceptual changes resulting from technological stimuli (Kordzadeh & Ghasemaghaei, 2021). For healthcare facilities and developers, fair, worker-centered shift scheduling supported by e-rostering that incorporates contextual factors results in enhanced job satisfaction and worker retention.

2 Background and Related Work

2.1 Healthcare Shift Scheduling

In healthcare shift scheduling, workers are assigned to specific shifts of various types (e.g., night or day shifts). This is a complex procedure because several requirements must be considered to uphold quality of care. These include operational requirements to satisfy the demand, such as a minimal number or skill mix of workers, as well as legal requirements, including minimum rest periods and rules for shift patterns (Erhard et al., 2018). These requirements constitute hard constraints that must be satisfied in every shift schedule; otherwise, the demand in a ward cannot be covered, or the schedule may violate labor laws. Requirements may vary depending on factors such as the type of organization and employment sector. For example, given persistent doctor shortages, doctors may exert greater bargaining power in accommodating scheduling preferences (Erhard et al., 2018). In addition, nursing homes typically apply a different skill mix than hospitals (Shin & Kim, 2024). Three main approaches for shift scheduling

have been established: cyclical scheduling, self-scheduling, and preference scheduling. Cyclical scheduling employs a fixed-shift rotation that ensures compliance with hard constraints. However, it is rarely used in practice because of its inflexibility in adapting to changing staff demands and worker preferences (Burke et al., 2004). Self-scheduling transfers scheduling responsibilities from supervisors to workers. However, this increases the complexity for workers and the risk of violating hard constraints (Wynendaele et al., 2021). Preference shift scheduling considers worker preferences while complying with hard constraints. However, manually creating monthly schedules is challenging and time-consuming for supervisors (Erhard et al., 2018). E-rostering tools are increasingly being used to efficiently manage the complexity of shift scheduling because they automatically generate (parts of) schedules that comply with requirements or support the detection of hard constraint violations (O'Connell et al., 2024; Petrovic, 2019). Recently, e-rostering has seen an increase in the use of artificial intelligence to improve the efficiency and flexibility of e-rostering by processing larger amounts of data (Renggli et al., 2025). However, artificial intelligence in e-rostering remains ethically and practically challenging due to the processing of sensitive personal data and the risk of biased training data (Gerlach et al., 2025).

Independent of the scheduling approach, recent research identified schedule predictability and control over schedules as the primary factors for worker-centered shift scheduling (Barker et al., 2025; Emmanuel et al., 2024). Predictable schedules are characterized by consistent shift patterns over multiple scheduling periods and minimize the number of last-minute changes (Emmanuel et al., 2024). If shift schedules lack predictability, shift types and working days vary for individual workers without an apparent pattern, making it difficult for them to plan and engage in their lives outside of work and resulting in adverse impacts on their well-being (Kim et al., 2025). However, creating predictable schedules requires forecasting staff demands and variability (Burke et al., 2004). Although some recent studies have provided initial insights into e-rostering tools and algorithms for forecasting demand using available data, this progress has yet to be applied in healthcare facilities (Fuchs et al., 2025; Hu et al., 2025). Thus, e-rostering currently facilitates schedule predictability by supporting consistency across multiple scheduling periods through tracking and displaying past schedules (Howard et al., 2020; Petrovic, 2019). Scheduling control enables workers to voice requests and preferences to (not) work on certain days, shifts, or shift types, which allows them to adjust schedules to private matters, such as childcare or health conditions (Barker et al., 2025). Scheduling approaches that incorporate flexibility, such as preference shift scheduling and self-scheduling, enable workers to participate in the scheduling process. E-rostering enables enhanced participation opportunities by providing features that allow for recording detailed requests and preferences of the workers as well as considering more of them while still complying with requirements (O'Connell et al., 2024; Schlicker et al., 2021).

2.2 Fairness Perceptions in Healthcare Shift Scheduling

Fairness in healthcare shift scheduling is an essential factor influencing the job satisfaction of healthcare workers (Epstein et al., 2023). Thus, most e-rostering tools incorporate fairness measures and consider shift preferences to fairly allocate shifts to workers. This approach falls under the technical perspective on fairness, which seeks to achieve fair outcomes by developing technical solutions. In isolation, this perspective lacks a comprehensive understanding of what constitutes a fair shift schedule based on the perceptions of healthcare workers (Dolata et al., 2022; Erhard et al., 2018). Neglecting the perceptions of those affected can result in unfair outcomes because developers of the tools may have a different notion of fairness due to a different context or background compared to that of the healthcare worker. Thus, achieving fair outcomes requires complementing the technical perspective with a social perspective that focuses on the perceptions of individuals in their organizational contexts (Dolata et al., 2022). Therefore, IS research has adopted OJT to examine the social perspective on fairness, specifically how emerging technologies within organizations influence fairness (Dolata et al., 2022). In this tradition, OJT has recently been embedded into the S-O-R framework as a systematic approach to describe how perceptions of fairness mediate the relationship between technology-related stimuli and the behavioral response of interest in a given context (Ochmann et al., 2024). As we aim to advance the social perspective, specifically the perspective of workers, on healthcare shift scheduling, we engage with

recent qualitative studies that focus on two main OJT dimensions: distributive and procedural fairness (Kordzadeh & Ghasemaghaei, 2021).

Distributive fairness focuses on individuals' perceptions of fairness regarding the allocation of outcomes, typically based on the norms of equity, equality, and need (Colquitt, 2001). First, *equity* assumes that workers compare their relative outcomes (e.g., shift schedules) and contributions (e.g., effort or quality of work) with those of their coworkers to form perceptions of fairness. Second, *equality* demands that outcomes are evenly distributed among workers, independent of their individual contributions. Third, *need* requires that outcomes be allocated according to individual needs, independent of contributions (Colquitt, 2012; Kordzadeh & Ghasemaghaei, 2021). In healthcare shift scheduling, distributive fairness primarily focuses on the allocation of unpopular shifts (e.g., during holidays) or shift types (e.g., typically night shifts) (Uhde et al., 2020). Recent studies on fairness in shift scheduling suggest that healthcare workers prioritize the need norm, considering individual needs, such as childcare responsibilities or preferences for specific shift types. However, fulfilling all individual needs is rarely possible because legal and operational requirements must still be satisfied. Instead, needs should be balanced between workers within a ward, without demanding strict equality due to substantial differences in needs, captured by the concept of "*balanced need*" (Bieri & Matt, 2025). In addition to strict equality, equity is perceived as unfair because work performance and personal traits should not affect shift allocation (Uhde et al., 2020). Still, many scheduling algorithms base their fairness notions on strict equality and equity, which diverges from worker fairness perceptions (Fuchs et al., 2025).

Procedural fairness refers to the perception of fairness held by the affected individuals regarding internal decision-making procedures. Initially, procedural fairness was evaluated by the amount of voice or participation workers had during procedures. Later, rules to ensure fair procedures were incorporated into the definition of procedural fairness, including the consistency and unbiased nature of procedures (Colquitt, 2001). In healthcare shift scheduling, worker participation or scheduling control is critical because (non-)involvement results in perceived procedural (un)fairness (Uhde et al., 2020). Participation in shift scheduling involves mechanisms that may be supported by e-rostering, enabling workers to voice their requests and preferences, and adjust published schedules to align their personal and professional commitments (Epstein et al., 2023). However, the optimal design of worker participation to enhance fairness perceptions remains unclear, as the highest level of participation with self-scheduling, which transfers the task of creating schedules to workers, is not only perceived as fair because of challenges in resolving over- and understaffing or manipulation by coworkers (Wynendaele et al., 2021). In addition to participation, fair shift scheduling procedures should rely on accurate information and be applied consistently across all workers (Bieri & Matt, 2025). E-rostering is perceived as more objective and less prone to bias. In contrast, human scheduling by supervisors is perceived as offering more opportunities to voice individual views during the scheduling process (Schlicker et al., 2021).

3 Conceptual Model and Hypotheses

We build on the recent integration of OJT into the S-O-R framework to develop our conceptual model, as it offers an established, structured, yet parsimonious approach for examining the relationships among a stimulus, perceptions of fairness or the organism, and context-specific behavioral responses (Kordzadeh & Ghasemaghaei, 2021). In contrast to objective fairness that is typically formalized as mathematical notions, this integration is designed to capture individual subjective beliefs, such as fairness and work attitudes (Ochmann et al., 2024). In our study, e-rostering serves as the stimulus or external cue, as it is not explicitly chosen by healthcare workers but rather mandated by the organization for which they work. The organism captures the internal cognitive processes or states of individuals, such as their perceptions of fairness, which in our case are referred to as scheduling fairness perceptions of healthcare workers, divided into distributive and procedural fairness. Subsequently, scheduling fairness perceptions are linked to behavioral responses reflected in job satisfaction, an essential determinant of worker turnover in the healthcare sector (Brewer et al., 2015; Colquitt, 2012). In response to calls to incorporate important contextual factors that may alter the perceptual basis of affected individuals (Kordzadeh & Ghasemaghaei, 2021), we integrated three main worker-centered factors in

shift scheduling: scheduling control, schedule predictability, and relationships of workers with supervisors. The relationships in the conceptual model (Figure 1) are discussed in the following sections.

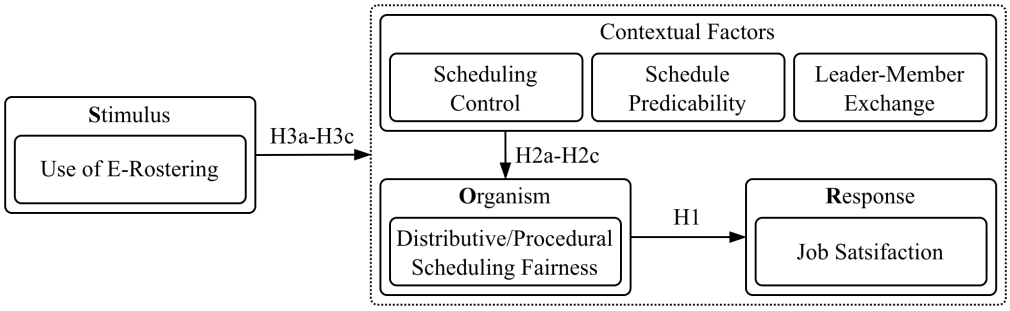

Figure 1. Conceptual model.

3.1 Scheduling Fairness and Job Satisfaction

A central topic in OJT is understanding how individuals form behavioral responses based on their perceptions of (un)fairness in the workplace. Previous research has consistently demonstrated that engaging in fairness benefits organizations. Workers who perceive fair treatment feel valued, resulting in higher levels of job satisfaction. Similarly, individuals who perceive unfairness tend to report lower job satisfaction (Colquitt, 2012; Schminke et al., 2000). The cognitive mechanism behind this effect is typically explained using social exchange theory. In contrast to economic exchange, social exchange involves the exchange of less tangible socio-emotional resources, such as fairness or work attitudes (Rupp & Cropanzano, 2002). The cognitive nature of social exchanges aligns with the S-O-R framework, which posits that the cognitive state of an individual influences the behavioral response (Colquitt, 2012; Ochmann et al., 2024). Specifically, the social exchange theory suggests that workers who receive fair treatment exhibit reciprocal behavior towards the organization, with attitudes and behaviors ultimately benefiting the organization (Colquitt, 2012). In shift scheduling, workers who receive fair scheduling procedures and shift schedules reciprocate with higher job satisfaction, ultimately benefiting the organization by decreasing turnover and enhancing work performance (Brewer et al., 2015; Rupp & Cropanzano, 2002). The social exchange theory is applicable when the relationship between the involved parties is long term, which is often the case with healthcare workers. Additionally, an exchange should entail elements of intrinsic relevance for both parties (Rupp & Cropanzano, 2002). Shift schedules are an integral part of the lives of healthcare workers, significantly affecting their ability to balance their personal and professional lives (Emmanuel et al., 2024). Healthcare facilities strive to retain workers. Thus, they benefit from increased job satisfaction, which results in lower turnover (Brewer et al., 2015). Although the effects of procedural and distributive fairness on job satisfaction may differ from reciprocal behavior, both are expected to be positive (Rupp & Cropanzano, 2002), resulting in the following hypothesis:

H1: Higher perceived scheduling fairness has a positive effect on perceived job satisfaction.

3.2 Contextual Factors and Scheduling Fairness

Previous studies established schedule predictability and scheduling control as essential factors in healthcare shift scheduling from the perspective of workers (Barker et al., 2025; Emmanuel et al., 2024). Predictable scheduling enables workers to arrange their private and professional lives without disruptions due to changing shift schedules. In contrast, inconsistent and unpredictable scheduling is characterized by frequent changes to the posted shift schedule with limited advance notice, hindering workers from aligning their personal lives outside of work with shift schedules (Emmanuel et al., 2024). The posted schedule forms a psychological contract that is not a formal, legally binding contract, as it cannot be enforced in a court of law. Instead, the worker perceives the promise of future intent by the organization (Morrison & Robinson, 1997). For workers, the announced schedule represents the organization's promise of a set number of working hours on specified days and times. If schedules are

unpredictable, organizations repeatedly fail to uphold their promises regarding posted schedules through (last-minute) changes, which workers perceive as breach of the psychological contract (Kim et al., 2025). As the psychological contract sets standards for perceived fairness, its breach causes feelings of distributive and procedural unfairness as the organization fails to meet these standards (Morrison & Robinson, 1997; Rosen et al., 2009). Psychological contract breaches foster the perceptions of workers that the scheduling procedure is inconsistent or biased and based on inaccurate information, thereby reducing procedural fairness perceptions (Bieri & Matt, 2025; Rosen et al., 2009). Disruptions to schedules hinder workers from reliably satisfying their needs beyond their professional lives, resulting in lower perceptions of distributive fairness (Barker et al., 2025; Kim et al., 2025). Thus, we propose the following hypothesis:

H2a: Higher perceived schedule predictability has a positive effect on perceived scheduling fairness.

Organizational justice literature has shown that when workers have opportunities to participate in decision-making processes, they perceive a greater sense of procedural fairness, even if the outcomes of those decisions are disadvantageous (Schminke et al., 2000). Consistent with this finding, recent research demonstrated that procedural fairness in healthcare shift scheduling is primarily influenced by the active involvement of workers in scheduling procedures (Barker et al., 2025; Uhde et al., 2020). Worker involvement, also referred to as scheduling control, ranges from simply voicing requests and preferences (participatory scheduling) to collaboratively creating a schedule with coworkers (self-scheduling). Therefore, if workers perceive greater control over shift schedules, they perceive greater procedural fairness (Uhde et al., 2020). In addition to its effects on procedural fairness, control over work schedules enables workers to actively align job demands with their individual needs and preferences. At minimum, scheduling control enables workers to record their requests and preferences, forming the basis for a later generation of schedules (Swanberg et al., 2011). Thus, scheduling control facilitates the incorporation of workers' needs into schedules as long as they align with operational and legal requirements, thereby enhancing distributive fairness (Bieri & Matt, 2025; Emmanuel et al., 2024). Therefore, we propose the following hypothesis:

H2b: Higher perceived scheduling control has a positive effect on perceived scheduling fairness.

Previous factors are primarily determined at the organizational level through formal policies, with limited influence from workers or their immediate supervisors (Epstein et al., 2023). However, the social exchange theory posits that fairness perceptions are affected not only by exchanges with the organization but also by interpersonal exchanges with supervisors (Colquitt, 2012). Thus, an essential determinant of fairness perception is the quality of the relationship between workers and supervisors, usually referred to as leader–member exchange (LMX) (Matta et al., 2024). Supervisors play a crucial role in shift scheduling as they balance operational and legal requirements with the needs of workers. They are often responsible for the final approval of the schedule and have broad discretion to grant special arrangements to consider the special needs of individual workers (Swanberg et al., 2011). Furthermore, through effective relationships with workers, supervisors gain a deeper understanding of the personal circumstances and shift preferences of workers, enabling them to create more tailored schedules or effectively address dissatisfaction with assigned shifts (Epstein et al., 2023). Accordingly, we expect high-quality LMX to be associated with more favorable perceptions of scheduling fairness. Similarly, low-quality LMX is likely to elicit perceptions of unfairness because workers with low-quality LMX are more likely to perceive the scheduling procedure as inconsistent or biased, such as by favoring coworkers over them (Dulebohn et al., 2012). In shift scheduling, this may occur when workers with high-quality LMX tend to receive preferential treatment (e.g., having greater input in the scheduling process) and more favorable final schedules by virtue of their status; that is, by being given enhanced consideration or awareness of their needs (Bieri & Matt, 2025). Accordingly, we hypothesize:

H2c: Higher perceived leader-member exchange has a positive effect on perceived scheduling fairness.

3.3 Comparison of E-Rostering and Human Scheduling

In the following, we discuss how stimulus e-rostering influences the relationships within the structural model compared to human scheduling, beginning with scheduling fairness and job satisfaction. An

argument in favor of e-rostering over human scheduling is the perceived objectivity and consistency of the underlying algorithm as well as its alleged immunity to the potential manipulation inherent in human decision-making (Ötting & Maier, 2018; Schlicker et al., 2021). Workers may consider e-rostering as a way to implement consistent and unbiased scheduling based on predefined rules. Thus, they perceive e-rostering as more trustworthy than human scheduling, which can be perceived as susceptible to influences from other workers, resulting in greater uncertainty regarding the fairness of the scheduling procedures (O'Connell et al., 2024; Ötting & Maier, 2018). Individuals facing greater uncertainty in the decision-making process place higher weight on procedural fairness as a determinant of job satisfaction (Diekmann et al., 2004). Compared with human scheduling, perceived certainty in e-rostering may diminish the role of procedural fairness in determining job satisfaction. Regarding distributive fairness, e-rostering offers features to accommodate the needs of workers more effectively into shift schedules by processing more input data about the needs in less time, while still ensuring compliance with legal and operational requirements (O'Connell et al., 2024; Petrovic, 2019). Because e-rostering can improve scheduling fairness compared to human scheduling by more effectively accounting for shift requests and preferences, the influence of distributive fairness on job satisfaction may be more pronounced when e-rostering is implemented (Howard et al., 2020). Thus, we hypothesize that scheduling fairness affects job satisfaction differently under e-rostering than under human scheduling:

H3a: Compared with human scheduling, the use of e-rostering strengthens the effect of distributive fairness and weakens the effect of procedural fairness on job satisfaction.

Regarding contextual factors, we expected a significant difference in the effects of scheduling control and LMX on scheduling fairness between e-rostering and human scheduling. From the perspective of workers, e-rostering offers greater opportunities for voice and participation by enabling features that allow workers to record their detailed individual needs. Additionally, increased processing capabilities in e-rostering enables workers to voice more requests and preferences overall (Petrovic, 2019; Schlicker et al., 2021). These enhanced opportunities for voicing requests and preferences should improve procedural fairness perceptions through increased scheduling control (Schminke et al., 2000; Uhde et al., 2020). Additionally, the enhanced recording of requests and preferences ultimately enables the integration of more worker needs into the final schedules, which results in enhanced distributive fairness (Bieri & Matt, 2025). The use of e-rostering may also impact the relationship between healthcare workers and their supervisors, as well as its effect on scheduling fairness. This is because e-rostering tools assume tasks previously performed by supervisors in the scheduling process, owing to their higher efficiency in processing requests and preferences compared to human scheduling (O'Connell et al., 2024). Consequently, supervisors may have a diminished role in scheduling procedures, resulting in a lower impact of LMX on the perceptions of shift schedules and scheduling procedures. Thus, we propose two hypotheses regarding the contextual factors when comparing e-rostering and human scheduling:

H3b: Compared with human scheduling, the use of e-rostering strengthens the effect of scheduling control on scheduling fairness.

H3c: Compared with human scheduling, the use of e-rostering weakens the effect of leader-member exchange on scheduling fairness.

4 Research Methodology

4.1 Measures and Scales

We used established scales from the organizational and management literature to measure our constructs and adapted them when necessary. All the constructs were reflective and analyzed using 7-point Likert scales (Table 1). Scheduling fairness perceptions were assessed using distributive and procedural fairness scales adapted from organizational justice research (Colquitt, 2001; Niehoff & Moorman, 1993). Contextual factors were measured using work schedule psychological contract breach for scheduling predictability (Kim et al., 2025), perceived scheduling control (Dwyer & Ganster, 1991), and LMX (Matta et al., 2024). Job satisfaction was measured according to the scale proposed by Mitchell et al. (2001). The scales for distributive fairness (Zahednezhad et al., 2021), procedural fairness (Schlicker

et al., 2021), LMX (Wynendaele et al., 2021), and job satisfaction (Mitchell et al., 2001) were previously applied in healthcare settings. The scales for scheduling control and schedule predictability were not used in healthcare settings, but they have been used in shift-scheduling contexts and therefore required no adjustments. To control for omitted variables, we collected several control variables, as suggested in previous literature (Epstein et al., 2023; Erhard et al., 2018; Ferris et al., 2008; Holland et al., 2019; Wynendaele et al., 2021): age, sex, region of residence, number of children, employment sector (i.e., doctor or nurse), experience at the current workplace, perceived workload, type of organization (e.g., hospital or nursing home), work pattern, group cohesion, and scheduling approach. Besides the scales for fairness perceptions (Maier et al., 2007) and LMX (Schyns & Paul, 2002), we translated the items into German using translation tools. Subsequently, we reviewed them extensively within the author team for comprehensibility.

Construct	**Likert Scale**	**#Items**	**Sample Item**
Job satisfaction (Mitchell et al., 2001)	1 (strongly disagree) – 7 (strongly agree)	3	In general, I like working here.
Distributive fairness (Niehoff & Moorman, 1993)	1 (strongly disagree) – 7 (strongly agree)	4	The shift schedules are fair.
Procedural fairness (Colquitt, 2001)	1 (never) – 7 (always)	6	Is the scheduling procedure free of bias across employees?
Schedule predictability (Kim et al., 2025)	1 (strongly disagree) – 7 (strongly agree)	5	So far, my employer has not done a great job of fulfilling its promises regarding my shift schedule to me.
Scheduling control (Dwyer & Ganster, 1991)	1 (none) – 7 (full)	3	How much control do you have over when you take vacations or days off?
Leader-member exchange (Matta et al., 2024)	1 (strongly disagree) – 7 (strongly agree)	7	My supervisor understands my problems and needs.

Table 1. Focal constructs and scales.

4.2 Sample

Variable		**Total**	**E-Rostering**		
			Yes	**No**	**P-Value**
N		185	83	102	
Age (years)		35.01	34.52	35.41	.565
Sex	Female	72.4%	68.7%	75.5%	.386
	Male	27.6%	31.3%	24.5%	
Employment Status	Full-Time	70.8%	71.1%	70.6%	1.000
	Part-Time	29.2%	28.9%	29.4%	
Employment Sector	Nurse	73.5%	73.5%	73.5%	1.000
	Doctor	26.5%	26.5%	26.5%	
Work Pattern	Rotating shifts, incl. night shifts	70.8%	71.1%	70.6%	1.000
	Rotating shifts without night shifts	29.2%	28.9%	29.4%	

Table 2. Selected demographics.

We recruited participants through the online sampling platform Prolific, a popular data-sampling platform in academia that has demonstrated high data quality compared with other platforms (Esch et al., 2025). We used the following prescreening criteria to collect a sample of healthcare workers affected by rotating shift work patterns: employment status (full-time or part-time), language fluency (English

or German), location of residency (North America or Europe), and employment sector (doctor or nurse). Furthermore, we applied custom prescreening to filter candidates who played a supervisory role in shift scheduling, as they may have a skewed perception of the schedules generated under their supervision. A priori, a sample of 155 observations was required to ensure 80% statistical power, assuming a 5% significance level and expected path coefficients greater than .11. A post hoc analysis using the inverse square root method estimated the required sample size to be 173 (Hair et al., 2024). Thus, the final sample of 185 participants who fully completed the survey exceeded both thresholds. The median completion time was approximately 12 minutes. Based on examples of e-rostering in the survey, 83 of the 185 participants reported that e-rostering was used in their wards. This group showed no significant differences in any prescreening criteria compared with the remaining participants (Table 2).

5 Data Analysis and Results

5.1 Structural Equation Model

To test our conceptual model, we estimated a structural equation model using partial least squares (PLS-SEM) with SmartPLS 4, which is suitable for examining extensions of established theories (Hair et al., 2024; Ringle et al., 2024). To ensure valid path and significance estimates for the hypothesis tests, we assessed the measurement and structural model (Wiener et al., 2023). We tested the measurement model using established practices and thresholds for construct reliability and validity, as well as discriminant validity indices (Hair et al., 2024; Table 3). Reliability was assessed through internal consistency measured using Cronbach's alpha (CA) and composite reliability (CR). All the constructs exceeded the suggested threshold of .7 for CA and CR. Convergent validity was confirmed because the average variance extracted (AVEs) for each construct exceeded the recommended threshold of .5. After removing one item from the reflective measure for procedural fairness with a loading of .51, the AVEs for all constructs significantly exceeded the recommended threshold at the 95% confidence level. We established discriminant validity by assessing the heterotrait–monotrait ratios (HTMT), which were all below .8, and therefore, lower than the established threshold of .85. Although the Fornell–Larcker criterion was met, as the square roots of the AVEs exceeded all inter-construct correlations, we refrained from declaring these values in Table 3 owing to its limitations in identifying discriminant validity issues (Hair et al., 2024).

Construct	Loadings	CA	CR	AVE	(1)	(2)	(3)	(4)	(5)	(6)
(1) SP	.907–.935	.943	.944	.853						
(2) SC	.737–.881	.768	.776	.687	.578					
(3) LMX	.683–.876	.922	.929	.683	.679	.594				
(4) DF	.842–.916	.912	.913	.792	.798	.737	.752			
(5) PF	.640–.831	.857	.862	.587	.729	.700	.662	.775		
(6) JS	.910–.922	.905	.907	.840	.537	.454	.604	.640	.598	
Note: SP = schedule predictability, SC = scheduling control, LMX = leader-member exchange, DF = distributive scheduling fairness, PF = procedural scheduling fairness, and JS = job satisfaction.										

Table 3. Reliability and validity criteria, and heterotrait–monotrait ratios.

The assessment of the structural model for multicollinearity was conducted according to established guidelines (Hair et al., 2024). As all variance inflation factors (VIF) were below three, there was no indication of multicollinearity. To rule out potential common method bias (CMB), we conducted five established tests (Wiener et al., 2023). First, the survey was designed using simple and focused questions, which were confirmed by positive participant feedback. Second, the correlations between the constructs were all less than .90. Third, Harman's single-factor test was performed using an exploratory factor analysis. The results showed that the first factor accounted for a maximum of 12% of the variance, which was below the established threshold of 50%. Fourth, as stated above, the assessment of

multicollinearity confirmed that the VIFs were below three for all constructs. Finally, a marker variable was added to the model, pointing to both independent and dependent constructs (Miller & Simmering, 2023). Including this marker variable did not alter the statistical significance or the size of the estimated path coefficients between the constructs. Therefore, CMB was not an issue in our analysis.

To evaluate the significance of the path coefficients, we conducted percentile bootstrapping with 10,000 subsamples assuming a significance level of 5% (Figure 2). The results showed that both distributive fairness (.227, p = .006) and procedural fairness (.189, p = .002) had positive and significant effects on job satisfaction. Regarding contextual factors, nearly all demonstrated significant positive relationships with fairness dimensions at the 5% level, except for the effect of LMX on procedural fairness (.079, p = .172). Specifically, LMX was significantly related to distributive fairness (.197, p = .006). Schedule predictability was significantly related to both distributive (.318, p < .001) and procedural fairness (.322, p < .001). The path coefficients between scheduling control and distributive fairness (.292, p < .001) and procedural fairness (.291, p < .001) were also positive and significant. Regarding the explanatory power, the model yielded R^2-values of .719 for distributive fairness, .582 for procedural fairness, and .492 for job satisfaction. Furthermore, all effect sizes for the significant paths ranged from .034 to .163, indicating a small to medium effect (Hair et al., 2024). The overall model showed a good fit, with an SRMR of .063, which was below the threshold of .08. Regarding predictive power assessed with PLSpredict, our model achieved $Q^2_{predict}$-values above zero for all endogenous constructs, suggesting a better fit than the indicator average. Furthermore, it performed better in terms of RMSE and MAE than the indicator average and linear model benchmarks for all endogenous constructs and indicators of these constructs, except for two indicators of procedural fairness. Thus, the model demonstrated better predictive power for most indicators of the endogenous constructs, indicating medium to large predictive power. Additionally, in the cross-validated predictive ability test (CVPAT), the model outperformed the benchmarks for all endogenous constructs, confirming its large predictive power (Hair et al., 2024).

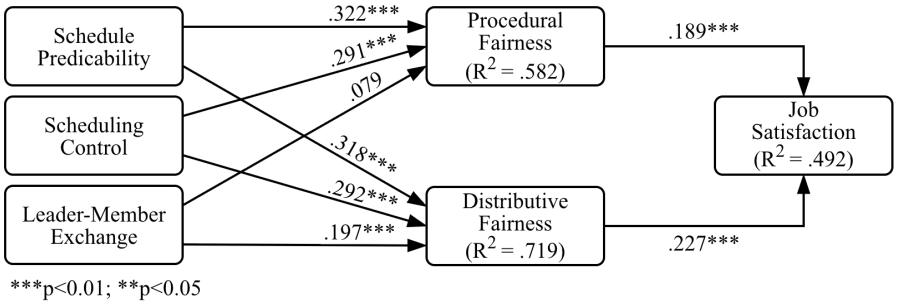

Figure 2. Structural model results.

Recent research has emphasized the importance of robustness checks in PLS-SEM, specifically addressing heterogeneity, endogeneity, and nonlinearity (Vaithilingam et al., 2024). To evaluate the unobserved heterogeneity, we ran finite mixture partial least squares (FIMIX-PLS) for up to three data segments (Table 4). For the two- and three-segment model, entropy statistics (EN) exceeded .5, indicating sufficient consistency. As suggested, we analyzed the AIC3 and CAIC jointly, as well as the BIC information criteria (Sarstedt et al., 2022). The assessment of different information criteria revealed that the two- and three-segment models did not outperform the single-segment model. We addressed endogeneity concerns by including control variables and performing a model comparison (MODCOM) with an alternative model that included two additional theory-grounded paths from LMX to job satisfaction (Rupp & Cropanzano, 2002) and scheduling control to schedule predictability (Kim et al., 2025). Except for two items of procedural fairness, the hypothesized model performed better in CVPAT and PLSpredict than the alternative model. Regarding BIC, the hypothesized model performed better in terms of job satisfaction but worse in terms of both fairness dimensions. In summary, MODCOM with the alternative model showed no considerable improvement compared to our more parsimonious hypothesized model. Similarly, to assess nonlinearity, we compared the hypothesized model with an alternative model that included quadratic relationships between all constructs. Again, except for two items of procedural fairness, the hypothesized model outperformed the alternative model in both the

CVPAT and PLSpredict. Regarding BIC, the hypothesized model performed better for all endogenous constructs. Therefore, based on MODCOMs, our model specification was adequate (Hair et al., 2024).

# of Segments	AIC3	CAIC	BIC	EN
1	1190	1332	1288	-
2	1200	1487	1398	.663
3	1170	1602	1468	.836

Table 4. FIMIX-PLS results.

5.2 Multigroup Analysis

To assess the impact of stimulus e-rostering on our structural model, we conducted a multigroup analysis (MGA) using SmartPLS 4 with an assumed significance level of 5% (Ringle et al., 2024). In PLS-SEM, a meaningful MGA requires establishing measurement invariance of composites (MICOM), which comprises three steps (Hair et al., 2024; Table 5): First, configural invariance was satisfied because we did not adjust the measurement model for any of the groups. Second, compositional invariance requires the correlations between the composite scores of both groups to be 1. After removing one item related to scheduling control, no significant differences were observed in the correlations. Third, the mean and variance for each construct should be equal across groups, which was fulfilled as no significant differences were detected. Because all criteria in MICOM were satisfied, we proceeded with MGA.

The permutation MGA, using 10,000 permutations, revealed multiple differences in path coefficients between e-rostering and human scheduling (Hair et al., 2024). For e-rostering, distributive fairness had a significantly more positive effect on job satisfaction (diff = .352, p = .033), whereas procedural fairness had a significantly weaker effect on job satisfaction than under human scheduling (diff = -.337, p = .007). Regarding contextual factors, scheduling control demonstrated a significantly more substantial effect on distributive fairness in the e-rostering group (diff = .265, p = .008), whereas its effect on procedural fairness did not differ significantly (diff = .105, p = .251). Schedule predictability showed no significant difference in its relationship with distributive fairness (diff = -.094, p = .253) but had a significantly stronger effect on procedural fairness in the e-rostering group (diff = .318, p = .038). Finally, LMX showed no significant group differences in its effects on either distributive fairness (diff = -.048, p = .388) or procedural fairness (diff = -.083, p = .326).

Construct	Configural Invariance	Compositional Invariance	Equal Means	Equal Variances
		Correlation (p-value)	Difference (p-value)	Difference (p-value)
(1) SP	Yes	1.000 (.941)	-.229 (.125)	.384 (.055)
(2) SC	Yes	.995 (.088)	.075 (.615)	-.222 (.255)
(3) LMX	Yes	.999 (.405)	-.123 (.419)	-.116 (.562)
(4) DF	Yes	1.000 (.231)	-.185 (.218)	.036 (.841)
(5) PF	Yes	.997 (.236)	-.122 (.415)	-.256 (.180)
(6) JS	Yes	1.000 (.332)	-.263 (.077)	.346 (.173)
Note: SP = schedule predictability, SC = scheduling control, LMX = leader-member exchange, DF = distributive scheduling fairness, PF = procedural scheduling fairness, and JS = job satisfaction.				

Table 5. MICOM results.

6 Discussion

Our structural model responds to three critical relations in e-rostering. First, we show how distributive and procedural scheduling fairness are formed by contextual factors. Second, we revealed how scheduling fairness dimensions affect job satisfaction. Third, we utilize MGA to determine how e-

rostering impacts the relationships within the structural model. Using data from 185 healthcare workers, the structural model satisfied all quality and robustness criteria, including CMB, hidden endogeneity, and model comparisons, thereby allowing valid tests of the hypotheses within the structural model (Hair et al., 2024). We obtained support for all hypotheses in the structural model, except for the partial support of H2c (Table 6), as LMX has a significant positive effect on distributive fairness but not on procedural fairness. Contrary to previous OJT literature, healthcare workers do not seem to associate their exchange with the supervisor with the (un)fairness of the scheduling procedure (Dulebohn et al., 2012). This finding suggests that lower LMX is not linked to inconsistent or biased scheduling procedures where supervisors favor selected, well-connected workers, nor is higher LMX connected with enhanced input or voice during scheduling procedures (Uhde et al., 2020). However, LMX still influences workers' perceptions of distributive scheduling fairness, as an enhanced exchange supports supervisors in understanding worker needs, enabling supervisors to adapt schedules more accurately to meet these needs (Epstein et al., 2023). Regarding the other two contextual factors, schedule predictability and scheduling control emerge as positive and strong predictors of scheduling fairness, aligning with prior qualitative research (Barker et al., 2025; Bieri & Matt, 2025). Although prior qualitative research has identified these factors, their relative importance has not yet been explicitly evaluated. The quantitative results indicate that both factors are equally important in determining scheduling fairness. Furthermore, we demonstrate that the nearly equal positive impacts of procedural and distributive scheduling fairness on job satisfaction remain even after controlling for several covariates. On a broader scale, our findings on scheduling fairness align with prior work on OJT, demonstrating that fairness perceptions are adequate predictors of work attitudes (Colquitt, 2012).

	Hypothesis	**Support**
H1	Higher perceived scheduling fairness has a positive effect on perceived job satisfaction.	Yes
H2a	Higher perceived schedule predictability has a positive effect on perceived scheduling fairness.	Yes
H2b	Higher perceived scheduling control has a positive effect on perceived scheduling fairness.	Yes
H2c	Higher perceived leader-member exchange has a positive effect on perceived scheduling fairness.	Partially

Table 6. Hypotheses structural model.

Regarding the MGA used to assess stimulus e-rostering (Table 7), we obtained support for H3a, which stated that distributive fairness has a more substantial effect on job satisfaction, whereas procedural fairness has a weaker effect. These findings align with previous arguments comparing algorithmic and human decision-making, suggesting that the underlying procedures in algorithms are perceived as more objective and consistent, thereby reducing the weight of procedural fairness (Ötting & Maier, 2018). Additionally, e-rostering offers an increased processing capacity to consider worker needs in the final schedules, which increases the impact of distributive fairness (Howard et al., 2020). Regarding contextual factors, scheduling control has stronger effects on both fairness dimensions under e-rostering than under human scheduling. This aligns with e-rostering features that record the detailed requests and preferences of workers, thereby enhancing the accommodation of their needs in the final schedules (Petrovic, 2019; Schlicker et al., 2021). However, only the effect on distributive fairness is significantly more substantial, resulting in a partial support for H3b. Furthermore, MGA did not reveal significant negative differences in the effects of LMX on either fairness dimension, thus failing to support H3c. The following two aspects may offset our hypothesized effect. First, current e-rostering tools are not yet autonomous in creating high-quality schedules and require supervisors to manually adjust them. An effective exchange between workers and supervisors ensures that supervisors remain informed about workers' needs, which helps them make final adjustments to their schedules (Epstein et al., 2023; Petrovic, 2019). Second, committed supervisors have been identified as key factors in the successful implementation of e-rostering (O'Connell et al., 2024). Thus, when supervisors are committed to e-

rostering and have strong ties to workers, they can positively affect the perceptual basis of fairness in e-rostering by highlighting its advantages (Wynendaele et al., 2021).

	Hypothesis	Support
H3a	Compared with human scheduling, the use of e-rostering strengthens the effect of distributive fairness and weakens the effect of procedural fairness on job satisfaction.	Yes
H3b	Compared with human scheduling, the use of e-rostering strengthens the effect of scheduling control on scheduling fairness.	Partially
H3c	Compared with human scheduling, the use of e-rostering weakens the effect of leader-member exchange on scheduling fairness.	No

Table 7. Hypotheses multigroup analysis.

7 Implications

This study contributes to the literature on OJT, healthcare shift scheduling, and e-rostering. Regarding OJT, we address calls to extend its integration into the S-O-R framework with contextual factors (Kordzadeh & Ghasemaghaei, 2021; Ochmann et al., 2024). Thus, our context-specific model explains how the organism (i.e., scheduling fairness) is shaped by contextual factors, and how both scheduling fairness dimensions influence the behavioral response, that is, job satisfaction. Regarding the stimulus, such as e-rostering, integrating contextual factors allows us to observe changes in the impact on the organism as well as to gain a deeper understanding of the mechanisms through which the stimulus induces perceptual changes in the organism's antecedents. For example, we observed that employing e-rostering strengthened the effect of scheduling controls on fairness perceptions. Furthermore, we follow a tradition in OJT research by analyzing the relative importance of distributive and procedural fairness in determining individual work attitudes (Colquitt, 2012). Although both scheduling fairness dimensions are critical for the overall job satisfaction of healthcare workers, we reveal a shift in salience from procedural to distributive fairness in e-rostering compared to human scheduling, which aligns with arguments about perceptual changes when introducing algorithmic decision-makers (Ötting & Maier, 2018). This finding has direct implications for research on fairness perceptions in fields with emerging technologies, as the introduction or replacement of technologies may alter the relative importance of fairness dimensions in determining work attitudes, resulting in different design recommendations, such as whether explanations should focus on procedures or outcomes. Since insights derived under prior decision-making arrangements may not generalize to new technologies, we emphasize examining the fairness implications of new algorithmic decision-makers, including e-rostering (Schlicker et al., 2021).

Regarding contextual factors, our findings emphasize the importance of scheduling control and schedule predictability as key determinants of scheduling fairness. Whereas scheduling control is embedded in e-rostering through participation mechanisms (Petrovic, 2019), e-rostering tools for forecasting staff demands to increase schedule predictability remain limited (Fuchs et al., 2025). However, disruptions that reduce predictability can undermine fairness, as healthcare workers struggle to balance their professional and personal lives. Thus, considerable potential lies in advancing the limited recent research on e-rostering that builds on advances in artificial intelligence and scheduling algorithms that utilize historical and real-time data on capacity utilization in healthcare facilities. Such approaches may close the gap in schedule predictability within e-rostering (Hu et al., 2025). However, while workers are willing to share sensitive data to enhance scheduling fairness, it is unclear whether this extends to other necessary data, such as capacity utilization data that include patient information (Bieri & Matt, 2025).

Our findings have practical implications for shift scheduling in healthcare facilities and software developers of e-rostering tools. We demonstrated the impact of distributive and procedural scheduling fairness on job satisfaction, which highlights a path for healthcare facilities to retain workers in the healthcare sector through fair shift scheduling (Brewer et al., 2015). When e-rostering is employed, distributive fairness carries more weight than procedural fairness in determining job satisfaction. Since distributive fairness primarily depends on the norm that determines how shifts are allocated to workers,

we stress a paradigm shift from the dominant strict equality norm, which conflicts with workers' perceptions, toward the (balanced) need norm that accounts for shift preferences and requests of all workers based on their individual professional and private circumstances (Bieri & Matt, 2025; Fuchs et al., 2025). The more pronounced role of distributive fairness in e-rostering also suggests that providing adequate explanations of final schedules is even more relevant. An enhanced understanding of schedules, such as the reasoning for choosing among conflicting requests of different workers, facilitates distributive fairness (Schlicker et al., 2021; Uhde et al., 2020). We show how contextual factors in shift scheduling influence scheduling fairness and how they change the perceptual basis of healthcare workers in e-rostering compared with human scheduling. Healthcare facilities without flexible work arrangements can enhance both dimensions of fairness by enabling scheduling control through established participation mechanisms in e-rostering tools (O'Connell et al., 2024). The flexibility provided by artificial intelligence may further promote scheduling control by enabling the processing of more scheduling preferences and requests (Gerlach et al., 2025). Despite these mechanisms in e-rostering tools, they currently do not provide features for adequately forecasting staff demand (Hu et al., 2025). However, the developers of these tools may have the opportunity to directly access real-world data from their clients to develop accurate forecasting features using artificial intelligence (Renggli et al., 2025). In the meantime, healthcare facilities can use e-rostering to achieve more consistent schedules across individuals and time periods by easily tracking past schedules, thereby enhancing schedule predictability (Howard et al., 2020). Regardless of whether e-rostering is used, the relationship between workers and supervisors is essential to the perception of distributive fairness. However, the role of supervisors is evolving depending on the context. In human scheduling, they have an essential understanding of workers' needs and manage dissatisfaction with shift schedules. In e-rostering, they also represent e-rostering and convince workers that e-rostering capabilities ultimately result in fairer shift schedules (Epstein et al., 2023; O'Connell et al., 2024).

8 Limitations, Future Research, and Conclusion

This study has several limitations. First, numerous contextual and individual factors may influence scheduling fairness and work attitudes (Colquitt, 2001; Wynendaele et al., 2021). Although we controlled for several variables suggested in the literature and examined the robustness of our hypothesized model, we cannot entirely rule out unobserved heterogeneity that could bias the estimated path coefficients. Second, the participants self-assessed the use of e-rostering in their wards, and the specific implementations potentially varied, which may have biased the MGA. As the introduction of e-rostering is rare because of its costs for healthcare facilities, future research should seize available opportunities to assess how the introduction of e-rostering affects the perceptions of workers within controlled real-world settings, with comparable e-rostering implementations and larger groups of participants (Erhard et al., 2018). In this regard, our sample, while sufficient under the pre- and post-hoc power analyses, remains limited in size given its demographic heterogeneity. Third, future research is required on the design of scheduling control and its interrelation with scheduling efficiency. Our results suggest that scheduling control enhances scheduling fairness, but the optimal design of scheduling control remains unclear. For example, allowing workers to submit more requests increases control, but also increases the risk of unmet requests. Similarly, self-scheduling is not only perceived as fair (Wynendaele et al., 2021). Furthermore, increasing scheduling control in combination with schedule predictability may necessitate that healthcare facilities relinquish some scheduling flexibility to adapt to changing staff demands and maintain payroll efficiency (Kim et al., 2025). This trade-off between flexibility for workers and facilities illustrates the interdependence of social and organizational aspects in healthcare shift scheduling, complicating the design of optimal shift scheduling (Epstein et al., 2023).

In summary, we advance the understanding of fairness in healthcare shift scheduling from the perspective of workers by developing and empirically testing a conceptual model that connects contextual factors, such as schedule predictability, scheduling control, and LMX, to scheduling fairness, and, in turn, to job satisfaction. By combining the conceptual model with our e-rostering analysis, we offer recommendations for worker-centered shift scheduling and e-rostering. Focusing on the perceptions of healthcare workers, we address a key challenge in the sector by identifying ways to

enhance job satisfaction through fair shift scheduling, thereby supporting worker retention (Brewer et al., 2015). Although our findings are primarily applicable to the healthcare sector owing to its unique characteristics, they may also inform research and shift scheduling in other sectors (Erhard et al., 2018).

References

Barker, H. R., Griffiths, P., & Dall'Ora, C. (2025). "I don't think there's necessarily a one size fits all" negotiating competing priorities in nurse shift scheduling: A qualitative study. *BMC Nursing*, *24*, 1048.

Bieri, M., & Matt, C. (2025). Algorithmic fairness in shift scheduling – Assessing the fairness perceptions of healthcare workers. *Proceedings of the 2025 Americas Conference on Information Systems*.

Brewer, C. S., Chao, Y.-Y., Colder, C. R., Kovner, C. T., & Chacko, T. P. (2015). A structural equation model of turnover for a longitudinal survey among early career registered nurses. *International Journal of Nursing Studies*, *52*(11), 1735–1745.

Burke, E. K., De Causmaecker, P., Berghe, G. V., & Van Landeghem, H. (2004). The state of the art of nurse rostering. *Journal of Scheduling*, *7*(6), 441–499.

Colquitt, J. A. (2001). On the dimensionality of organizational justice: A construct validation of a measure. *Journal of Applied Psychology*, *86*(3), 386–400.

Colquitt, J. A. (2012). Organizational justice. In S. W. J. Kozlowski (Ed.), *The Oxford Handbook of Organizational Psychology* (Vol. 1, pp. 526–547). Oxford University Press.

De Bienassis, K., & Lafortune, G. (2024). Are working environments for healthcare workers improving? *OECD*.

Diekmann, K. A., Barsness, Z. I., & Sondak, H. (2004). Uncertainty, fairness perceptions, and job satisfaction: A field study. *Social Justice Research*, *17*(3), 237–255.

Dolata, M., Feuerriegel, S., & Schwabe, G. (2022). A sociotechnical view of algorithmic fairness. *Information Systems Journal*, *32*(4), 754–818.

Dulebohn, J. H., Bommer, W. H., Liden, R. C., Brouer, R. L., & Ferris, G. R. (2012). A meta-analysis of antecedents and consequences of leader-member exchange: Integrating the past with an eye toward the future. *Journal of Management*, *38*(6), 1715–1759.

Dwyer, D. J., & Ganster, D. C. (1991). The effects of job demands and control on employee attendance and satisfaction. *Journal of Organizational Behavior*, *12*(7), 595–608.

Emmanuel, T., Griffiths, P., Lamas-Fernandez, C., Ejebu, O.-Z., & Dall'Ora, C. (2024). The important factors nurses consider when choosing shift patterns: A cross-sectional study. *Journal of Clinical Nursing*, *33*(3), 998–1011.

Epstein, M., Arakelian, E., Tucker, P., & Dahlgren, A. (2023). Managing sustainable working hours within participatory working time scheduling for nurses and assistant nurses: A qualitative interview study with managers and staffing assistants. *Journal of Nursing Management*, *2023*, 8096034.

Erhard, M., Schoenfelder, J., Fügener, A., & Brunner, J. O. (2018). State of the art in physician scheduling. *European Journal of Operational Research*, *265*(1), 1–18.

Esch, D. T., Mylonopoulos, N., & Theoharakis, V. (2025). Evaluating mobile-based data collection for crowdsourcing behavioral research. *Behavior Research Methods*, *57*, 106.

Ferris, D. L., Brown, D. J., Berry, J. W., & Lian, H. (2008). The development and validation of the workplace ostracism scale. *Journal of Applied Psychology*, *93*(6), 1348–1366.

Fuchs, G., Schimmelpfeng, K., & Brunner, J. O. (2025). A roadmap for integrating fairness in personnel planning and scheduling in hospitals. *Operations Research, Data Analytics and Logistics*, *45*, 200479.

Gerlach, M., Renggli, F. J., Bieri, J. S., Sariyar, M., & Golz, C. (2025). Exploring nurse perspectives on AI-based shift scheduling for fairness, transparency, and work-life balance. *BMC Nursing*, *24*(1), 1161.

Hair, J. F., Hult, G. T. M., Ringle, C. M., Sarstedt, M., Richter, N. F., & Hauff, S. (2024). *Partial Least Squares Strukturgleichungsmodellierung: Eine Anwendungsorientierte Einführung* (2nd ed.). Verlag Franz Vahlen.

Holland, P., Tham, T. L., Sheehan, C., & Cooper, B. (2019). The impact of perceived workload on nurse satisfaction with work-life balance and intention to leave the occupation. *Applied Nursing Research*, *49*, 70–76.

Howard, F. M., Gao, C. A., & Sankey, C. (2020). Implementation of an automated scheduling tool improves schedule quality and resident satisfaction. *PLoS ONE*, *15*(8).

Hu, Y., Chan, C. W., & Dong, J. (2025). Prediction-driven surge planning with application to emergency department nurse staffing. *Management Science*, *71*(3), 2079–2126.

Kim, D., Hu, D., & Harold, C. M. (2025). Working around unpredictable clocks: Examining the impact of last-minute schedule changes on perceived contract breach and job performance. *Human Relations*, *78*(12), 1543–1570.

Kordzadeh, N., & Ghasemaghaei, M. (2021). Algorithmic bias: Review, synthesis, and future research directions. *European Journal of Information Systems*, *31*(3), 388–409.

Kumar, P., Holt, T., Wong, Y., & Kimeu, M. (2025). Heartbeat of health: Reimagining the healthcare workforce of the future. *McKinsey Health Institute*.

Maier, G. W., Streicher, B., Jonas, E., & Woschée, R. (2007). Gerechtigkeitseinschätzungen in Organisationen. *Diagnostica*, *53*(2), 97–108.

Matta, F. K., Frank, E. L., & Muir (Zapata), C. P. (2024). "Seeing eye to eye" about our relationship is good for us and everyone else: An examination of LMX agreement and views of fair treatment. *Organization Science*, *35*(4), 1489–1511.

Miller, B. K., & Simmering, M. J. (2023). Attitude toward the color blue: An ideal marker variable. *Organizational Research Methods*, *26*(3), 409–440.

Mitchell, T. R., Holtom, B. C., Lee, T. W., Sablynski, C. J., & Erez, M. (2001). Why people stay: Using job embeddedness to predict voluntary turnover. *Academy of Management Journal*, *44*(6), 1102–1121.

Morrison, E. W., & Robinson, S. L. (1997). When employees feel betrayed: A model of how psychological contract violation develops. *Academy of Management Review*, *22*(1), 226–256.

Niehoff, B. P., & Moorman, R. H. (1993). Justice as a mediator of the relationship between methods of monitoring and organizational citizenship behavior. *Academy of Management Journal*, *36*(3), 527–556.

Ochmann, J., Michels, L., Tiefenbeck, V., Maier, C., & Laumer, S. (2024). Perceived algorithmic fairness: An empirical study of transparency and anthropomorphism in algorithmic recruiting. *Information Systems Journal*, *34*(2), 384–414.

O'Connell, M., Barry, J., Hartigan, I., Cornally, N., & Saab, M. M. (2024). The impact of electronic and self-rostering systems on healthcare organisations and healthcare workers: A mixed-method systematic review. *Journal of Clinical Nursing*, *33*, 2374–2387.

OECD. (2025). *Health at a Glance 2025: OECD Indicators*.

Ötting, S. K., & Maier, G. W. (2018). The importance of procedural justice in human–machine interactions: Intelligent systems as new decision agents in organizations. *Computers in Human Behavior*, *89*, 27–39.

Petrovic, S. (2019). "You have to get wet to learn how to swim" applied to bridging the gap between research into personnel scheduling and its implementation in practice. *Annals of Operations Research*, *275*(1), 161–179.

Renggli, F. J., Gerlach, M., Bieri, J. S., Golz, C., & Sariyar, M. (2025). Integrating nurse preferences into AI-based scheduling systems: Qualitative study. *JMIR Formative Research*, *9*, e67747.

Ringle, C. M., Wende, S., & Becker, J.-M. (2024). *SmartPLS 4*. SmartPLS.

Rosen, C. C., Chang, C.-H., Johnson, R. E., & Levy, P. E. (2009). Perceptions of the organizational context and psychological contract breach: Assessing competing perspectives. *Organizational Behavior and Human Decision Processes*, *108*(2), 202–217.

Rupp, D. E., & Cropanzano, R. (2002). The mediating effects of social exchange relationships in predicting workplace outcomes from multifoci organizational justice. *Organizational Behavior and Human Decision Processes*, *89*(1), 925–946.

Sarstedt, M., Radomir, L., Moisescu, O. I., & Ringle, C. M. (2022). Latent class analysis in PLS-SEM: A review and recommendations for future applications. *Journal of Business Research*, *138*, 398–407.

Schlicker, N., Langer, M., Ötting, S. K., Baum, K., König, C. J., & Wallach, D. (2021). What to expect from opening up 'black boxes'? Comparing perceptions of justice between human and automated agents. *Computers in Human Behavior*, *122*, 106837.

Schminke, M., Ambrose, M. L., & Cropanzano, R. S. (2000). Effects of organizational structure on perceptions of procedural justice. *Journal of Applied Psychology*, *85*(2), 294–304.

Schyns, B., & Paul, T. (2002). Überprüfung einer deutschsprachigen Skala zum Leader-Member-Exchange-Ansatz. *Zeitschrift Für Differentielle Und Diagnostische Psychologie*, *23*(2).

Shin, J. H., & Kim, H. J. (2024). Comparison of nursing home workforce with acute-care setting nursing workforce: Using a national sample survey. *Geriatric Nursing*, *60*, 699–707.

Swanberg, J. E., McKechnie, S. P., Ojha, M. U., & James, J. B. (2011). Schedule control, supervisor support and work engagement: A winning combination for workers in hourly jobs? *Journal of Vocational Behavior*, *79*(3), 613–624.

Uhde, A., Schlicker, N., Wallach, D. P., & Hassenzahl, M. (2020). Fairness and decision-making in collaborative shift scheduling systems. *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, 527.

Vaithilingam, S., Ong, C. S., Moisescu, O. I., & Nair, M. S. (2024). Robustness checks in PLS-SEM: A review of recent practices and recommendations for future applications in business research. *Journal of Business Research*, *173*, 114465.

Wiener, M., Cram, W. A., & Benlian, A. (2023). Algorithmic control and gig workers: A legitimacy perspective of Uber drivers. *European Journal of Information Systems*, *32*(3), 485–507.

Wynendaele, H., Gemmel, P., Peeters, E., Myny, D., & Trybou, J. (2021). The effect of self-scheduling on organizational justice and work attitudes through leader-member exchange: A cross-sectional study using propensity scores. *International Journal of Nursing Studies*, *122*, 104032.

Zahednezhad, H., Hoseini, M. A., Ebadi, A., Farokhnezhad Afshar, P., & Ghanei Gheshlagh, R. (2021). Investigating the relationship between organizational justice, job satisfaction, and intention to leave the nursing profession: A cross-sectional study. *Journal of Advanced Nursing*, *77*(4), 1741–1750.